THE TENSOR PRODUCT OF TWO FLAT MODULES IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span's argument is self-contained (displayed associativity identity plus the composition-of-exact-functors principle); no external lemma is cited inside it.

Allowed standard prerequisites (editorial).

- Flatness means the tensor functor is exact. Stacks Project, Modules on Sites, definition of flat module. Supplies that \mathcal{G} and \mathcal{F} flat makes \mathcal{G} \otimes - and \mathcal{F} \otimes - exact.
- Associativity of the tensor product of modules. Stacks Project, Algebra, section on tensor products. Underlies the displayed identity used to view the composite as tensoring with \mathcal{G} \otimes \mathcal{F}.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. If $\mathcal{G}$ and $\mathcal{F}$ are flat $\mathcal{O}$-modules, then $\mathcal{G} \otimes_{\mathcal{O}} \mathcal{F}$ is a flat $\mathcal{O}$-module.*

Proof. This is true because

$$(\mathcal{G} \otimes_{\mathcal{O}} \mathcal{F}) \otimes_{\mathcal{O}} \mathcal{H} = \mathcal{G} \otimes_{\mathcal{O}} (\mathcal{F} \otimes_{\mathcal{O}} \mathcal{H})$$

and a composition of exact functors is exact. □

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